POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2018
Programme: BE | Full Marks: 100
Course: Digital Signal and Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Signal, System and Signal Processing. Explain the limitations of DSP. 7 1
   b) What is LTI system? Obtain the necessary and sufficient condition for causality of LTI system. 8 2
2. a) State and prove time shifting property of DTFS. 7 4
   b) Define z-transform. Determine the z-transform and ROC of the signal $x[n] = nan\ u[n]$. 8 3
3. a) Define DFT. Determine the FFT of the signal 8 4
   $x[n] = u[n] - u[n-4]$
   using DIT-FFT algorithm.
   b) Determine the Circular convolution of the signals 7 3
   $X1[n] = \{1,2,2,1\}$ and $x2[n] = \{2,1,1,2\}$
4. a) A certain discrete-time filter has the following data: 7 5
   Poles are at 0.2 and 0.4.
   Zeros are at -0.4 and origin.
   Gain of filter is 5.
   Determine cascade form realization
   b) A system has an impulse response 8 5
   $h[n] = (0.5)nu[n] + n(0.2)n\ u[n]$
   Determine parallel form realization.
5. a) Design a normalized linear phase FIR filter having the phase delay of $\tau = 4$ and at least 40 dB attenuation in the stop band. Also, obtain the magnitude /frequency response of the filter. 8 6
   b) Explain the FIR filter design by Kaiser Window. 7 6.3
6. a) If $H(s) = 1/[(s+1)(s+2)]$, find the corresponding $H(z)$ using IIM method for sampling frequency of 5 samples/sec. 7
   b) Explain the design of IIR filter using IIM method. Also write the limitations of IIM method. 8 7

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7. Write short notes on: **(Any two)**

a) Sampling of analog signals
b) Recursive and non-recursive systems
c) Radix-2 FFT algorithm

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